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Artist Talk: Hands On Stage: An Interface for Sound and Image Improvisation

Abstract

Hands On Stage is a performance interface designed from a percussionist's perspective. It explores the inter-relationship between the performer's hand movements, real-time video tracking techniques, and the recording and processing of sound in an integrated configuration. This paper describes the concept, interface design, technical development, and the creative approach to the visual and sound elements. The paper also discusses the project from a personal performance perspective.

Introduction

The intention of the project was to seek new performance approaches that would sit between two practices, percussion performance and new media art in a concert performance context. The initial research for the project has made me realise that many current experimental and new media concerts involve the integration of audio and visual domains, while the world of percussion encompasses a wide variety of performance practices and an extensive range of instruments. I decided to combine the two practices and build my own instrument, which would take percussion performance practices into another realm where the performer could have control over audio and visual domains with hand movements.

Hands On Stage was designed to be used as a tool to explore the relationship between the performer's movements, sounds produced by the performer and projected images. This interface has enabled the communication and interaction between three domains: hand movement, sound, and visual elements. In the aspect of sound design in relation to hand movement, the objective was to move beyond deterministic results and explore ways of controlling and mixing multiple sound elements in a real-time performance context. For the visual aspect, the projected images are simply captured and processed with video effects and projected in real-time.

Hands On Stage is a table-like interface. In creating the interface a number of areas, including human computer interface design, motion tracking, mapping, and contemporary performance practice were explored. Recent projects of interfaces designed for controlling and processing sound and/or image have included the *reacTable* and the *Morphing Table*, both designed for collaborative multi-user music making. There is also an interactive audiovisual installation, *Sound Rose*, which consists of a touch sensitive table and a projector. While these projects may use purpose-built table-like interface, the technology and the objectives are different.

This paper begins with an overview of the conceptual and technical development of *Hands On Stage*, and followed by a description of the configuration of sound and visual elements for an improvisation titled, *Untitled #1*. Finally this paper ends with a discussion of the project's future directions.

Hands On Stage

This section discusses the creation and the development of the interface, *Hands On Stage*.

Concept

The initial purpose of the project was to explore and enable the interaction between performer's gesture, and real-time sound and image processing to form an audiovisual performance approach from a percussionist's perspective.

Early experimental stages included looking for suitable materials and technologies in order to construct a table interface. Although sensor technologies could probably offer a more precise response to the performer's actions, it was my intention to simplify this part of progress. More time was spent in exploring the performer's hand movements as a control to sonic and visual outcomes. As a result, motion-tracking technology was applied instead of sensor technology.

The framework of the interface design was inspired by Thierry de Mey's composition *Musique de Table* for three percussionists who produce sounds by striking a table in many ways with their hands. Thus, *Hands On Stage* is constructed as a table-like interface enables the performer to have control over sets of configured parameters of the designed synthesis, patterns manipulation, and visual effects using hand movements. The input data for these parameters are analysed from the performer's hand movements, the percussive strikes of the table surface, and the activation of the buttons on the Wiimote controller.

Interface Construction

The construction of the *Hands On Stage* interface is rather simple. It is made of plywood pieces, a piece of Perspex covered with a sheet of tracing paper, two contact microphones, and a web camera.

Five pieces of plywood are used as the body and the bottom part of the table. Perspex is placed at the tabletop surface, and it is considered as the centre performing stage for the hands. The purpose of using a sheet of tracing paper is to diffuse the light. Two contact microphones are placed at two corners of the tabletop and these are connected to a M-Audio FireWire recording interface. A web camera is placed underneath the tabletop surface facing at tabletop for motion tracking purposes. However, one issue occurred while using an *I-Sight* camera as the choice of the web camera. Because *I-Sight* has a built-in microphone and it tends to reset the computer's audio input setting. As a result, the M-Audio recording interface crashes when there is an audio input conflict. This problem was resolved by replacing the camera to an *iMage* camera, which does not have a built-in microphone.

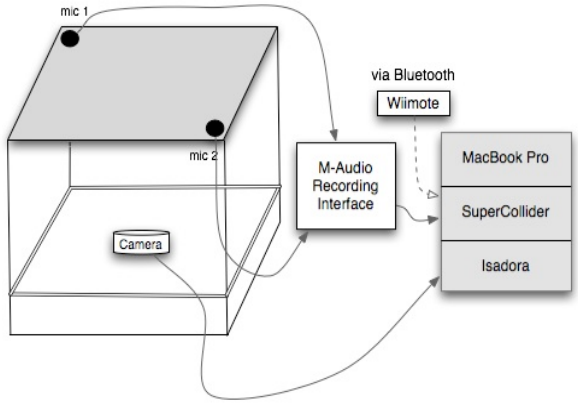


***Figure 1**. The table interface construction diagram.*

After constructing the physical interface, I soon realised there is one major issue in this interface. That is the lack of control over two microphones and a camera. Therefore, a Wiimote controller is added as the extension of the interface and its main function is to act as gates that allow or block the motion tracking data mapped to the sound parameters.

Technical Development

The interface technical development is categorised into three stages, generating input data from performer's action, analysis of the data, mapping of the parameters. In creating a personal approach towards audiovisual performance practices, the project eventually led to an integrated configuration through the use of two software programs, and *Isadora* running on one MacBook Pro. Also, as mentioned earlier, a Wiimote controller was included as a part of the interface.

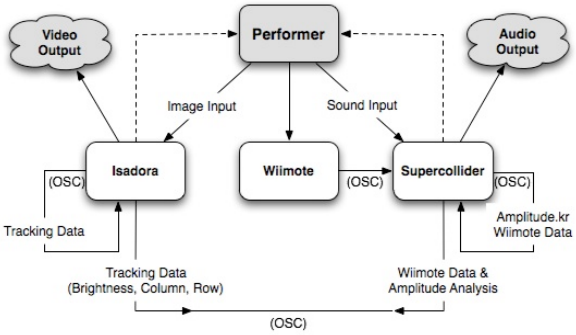


***Figure 2.** The diagram of the abstract relationship involves in utilising the Hands On Stage interface*

The sound elements, such as synth definitions, pattern-loops, and the amplitude analysis, are created and processed in *SuperCollider*. The amplitude analysis is used to trigger event when the amplitude value detected is above a certain threshold. It requires using two Ugens, a SynthDef and an OSCpathResponder.

```
SynthDef(\mic1, {|trigID=1|
//pathID=1
SendTrig.kr(Impulse.kr(40), trigID,
Lin-
Lin.kr(Amplitude.kr(AudioIn.ar(1)),
0.001, 0.5, 1, 20) );
}).play;

   OSCpathRespon-
der(s.addr,['/tr',nil,1],   {   |t,r,
msg| var threshold=20;
   m = msg[3].round(1).postln;
   if ( m > threshold, {
   s.sendMsg("/s_new",        \recBuf,
s.nextNodeID,    0,    1,    \bufnum,
~recBufnum.next);
   })
   }).add;
```

The image effects and video tracking analysis are done in *Isadora.* For video tracking analysis, a few Actors including Eyes and Threshold were used in *Isadora* in order to track the position of the shadow produced by the performer's hand. Also, an OSC Listener and an OSC Transmit is used here for receiving and sending OSC messages from and to *SuperCollider*. Here is an example patch:

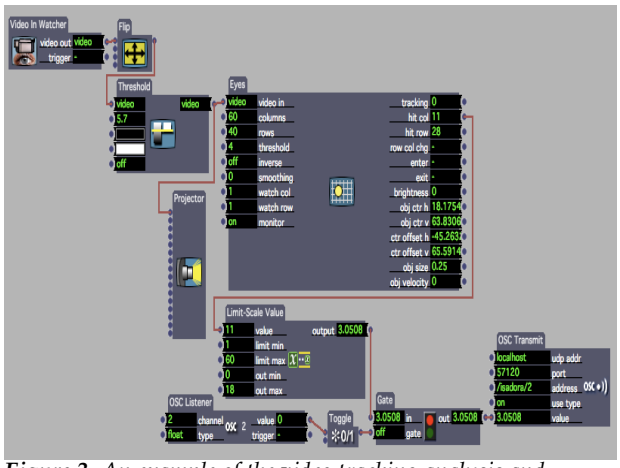


***Figure 3**. An example of the video tracking analysis and OSC data transmit in a graphic programming environment, Isadora.*

Because both programs support OSC (Open Sound Control) protocol, data communication is enabled between the two programs. For example, the performer's hand position is analysed in *Isadora*, and the values of the hand positions are passed onto the sound parameters in *SuperCollider* via OSC, when the gate function is called by the activation of the Wiimote buttons.

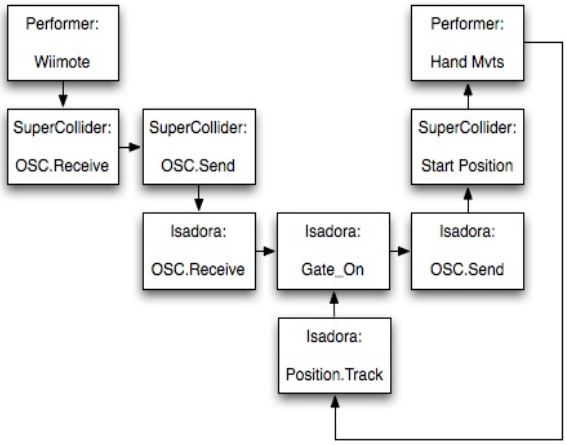


***Figure 4**. An example of the process of the communication between the performer, SuperCollider, and Isadora.*

Improvisation: Untitled # 1

The configuration for this sound and image improvisation, *Untitled # 1*, was first created and performed in 2007 at the CNMA's annual event, DUST. A revised version of this configuration was done in 2008. *Untitled # 1* is a structured improvisation, not a composition. Due to the nature of the mapping decisions and parameters, the result of the performance sounded quite different every time; thus the idea was to create an abstract plan for the improvisation. In fact, this idea suited the objective of the *Hands On Stage* interface, which aimed to include generating sounds that are beyond obvious results.



***Figure 5**. An abstract idea plotted for the improvisation: Untitled # 1.*

Untitled # 1 consists of three groups of sounds: percussive sounds, bell-like sounds, and white noises. The improvisation began real-time recorded percussive sounds. The percussive sounds are produced through striking the tabletop by the performer's hands. These sounds are recorded as sound files and loaded into the buffer, which execute a pattern as the code below, when called. Parameters in the Pbindef class, such as the start position (\startpos) and the buffer rate (\bufrate) are mapped to the video tracking data of the hand's position. The pattern used in the improvisation is a loop of a four-voiced pattern, which is produced by utilising the same sound file playing with different buffer rates and starting positions.

```
~loop1= Pbindef(\loop1,
\instrument, \playBuf,
\bufnum, nil,
\startpos, 0,
\out, 0,
\dur, Pseq([1, 1, 0.5, 0.5,
0.25],inf),
\amp, 0,
\bufrate, 1
);

~loop2= Pbindef(\loop2,
\instrument, \playBuf,
\bufnum, nil,
\startpos, 4, //different startpos
\out, 1,
\dur, Pseq([0.5, 0.5, 1, 1,
0.75],inf),
\amp, 0,
\bufrate, 2 //different bufrate
);
```

A graphic user interface (GUI) window is also created for this improvisation performance. The purpose of the creation of the GUI is to provide feedbacks of the status of the Wiimote button. Because they are used as on and off buttons, the colour of the GUI buttons changes in according to the Wiimote status.

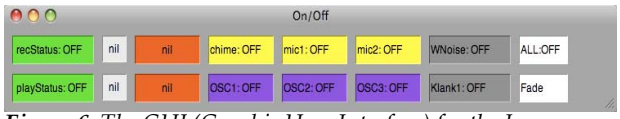


***Figure 6**. The GUI (Graphic User Interface) for the Improvisation: Untitled # 1.*

As for the visual aspects, three effects are designed for this improvisation. The web camera not only does the video tracking analysis, but it

also captures the images of the shadow movements on the tabletop, and processes them in real-time in *Isadora*.



***Figure 7**. An image of projected visuals from the rehearsal of Untitled # 1.*

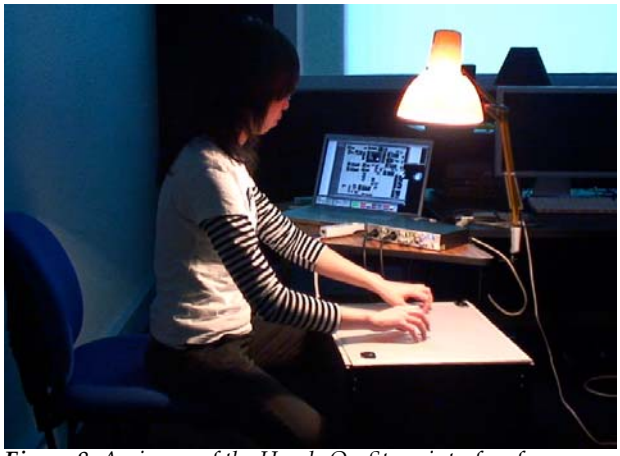


***Figure 8**. An image of the Hands On Stage interface from the rehearsal of Untitled # 1.*

Conclusion

Utilising the *Hands On Stage* interface for a sound and image performance is a challenging task. First, there is the technical knowledge required, in order to produce and have control over the sounds and image effects. Second, a major distinction between performing this experimental instrument and the conventional instruments is the constant involvement in listening, watching the processed visuals, and reading input data is always required while using such interface. It is perhaps not an expressive musical instrument, because much more attention was needed in order to gain more control over sound and image processing.

Future directions include exploring various configurations and approaches toward a theatrical performance, as well as implementing the technologies, designed for the *Hands On Stage*, onto a different surface, such as a drum skin, to resolve the limitation of the sounds produced from the striking of the tabletop.

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